

An Updated Deterministic-Envelope/Quotient Certificate Ledger

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Abstract

We consolidate RH-262–RH-269 into a logically explicit certificate ledger. The deterministic target now has a certified boundary constant, an exact all-order parity dictionary, a direct order-29 logarithmic tail below 2.6624745×10^{-5} , a unified envelope $|a_n| < 48q_*^n$, and the sharp law $a_n/q_*^n \rightarrow 1$. In contrast, the archive supplies no legal anchored head, cloud coefficient bridge, or uniform quotient tail; the sufficient continuum quotient criterion has none of its four hypotheses verified. Thus the five-component vector is $(0, 0, 0, 1, 1)$, with exactly two satisfied obligations and zero complete certificates in the audited route. The zero count is scoped bookkeeping, not a global nonexistence theorem.

1 Objects and certificate semantics

Let the Hardy-scaled deterministic target be normalized by

$$\log G_H(z) = - \sum_{n \geq 2} \frac{a_n}{n} z^n, \quad r_H = \frac{17}{20}. \quad (1)$$

For a selected moving-cloud or quotient family, write its candidate coefficients as $\tau_{n,s}$. A complete head–tail certificate requires more than a target estimate: it must identify a legal finite head, prove that its coefficients match the deterministic target in the stated protocol, and control both quotient and target tails uniformly.

Definition 1.1 (Five-obligation ledger). Use the fixed order (H, B, Q, A, M) , where

$$\begin{aligned} H &= \text{legal anchored head}, & B &= \text{coefficient bridge}, \\ Q &= \text{uniform quotient tail}, & A &= \text{analytic target tail}, \\ M &= \text{certified target boundary constant}. \end{aligned} \quad (2)$$

the complete-certificate predicate is the conjunction $H \wedge B \wedge Q \wedge A \wedge M$.

The five entries are deliberately independent. In particular, a theorem about the deterministic sequence a_n cannot by itself establish that a noise-dependent spectral selection produces those coefficients. This is the same firewall that separated the deterministic target from the moving-cloud trace-envelope question in RH-241 [3].

2 The deterministic target stack

The target side now consists of exact analytic identities and certified interval bounds rather than a finite root-rate extrapolation. RH-262 proves, using the RH-15 factorization and the RH-13 Arb certificate [1, 2, 7],

$$M_{7/5} := \sup_{|z|=7/5} |\log G_H(z)| < 107.906078 < 108. \quad (3)$$

RH-263 then gives the all-order parity dictionary [8]: for odd $n \geq 3$,

$$a_n = \frac{(r_H \lambda)^{-n}}{1 + \lambda^{-n}}, \quad (4)$$

while for $k \geq 1$,

$$a_{2k} = r_H^{-2k} \left[2 \operatorname{tr}(T^k) + \frac{2\lambda^{-2k}}{1 + \lambda^{-k}} - \frac{\lambda^{-2k}}{1 - \lambda^{-2k}} \right]. \quad (5)$$

The comparison against the 27 archived RH-253 rows through order 28 has maximum floating residual $6.442918843840156 \times 10^{-14}$; the proof of (4)–(5) is factorwise and does not depend on that finite check [5].

Put

$$q_* = (r_H \lambda)^{-1} = 0.7008752258547757 \dots, \quad \rho_* = q_*^{-1} = 1.4267874838640739 \dots \quad (6)$$

RH-267 proves the unified all-order envelope

$$|a_n| < 48q_*^n \quad (n \geq 2), \quad (7)$$

and RH-268 proves its base is sharp in the deterministic sense [12, 13]:

$$\frac{a_n}{q_*^n} \longrightarrow 1. \quad (8)$$

Thus no bound $|a_n| \leq Cq^n$ can hold for every n with $q < q_*$, although the prefactor 48 is not claimed to be optimal.

Proposition 2.1 (Envelope-only logarithmic tail). *If $0 \leq R < \rho_*$ and $N \geq 2$, then (7) implies*

$$\sum_{n \geq N} \frac{|a_n| R^n}{n} < \frac{48(q_* R)^N}{N(1 - q_* R)}. \quad (9)$$

At $R = 1$ and $N = 29$, the certified comparison $q_* < 0.700876$ gives the clean bound

$$\sum_{n \geq 29} \frac{|a_n|}{n} < 0.000184751. \quad (10)$$

Proof. For $n \geq N$, replace $1/n$ by $1/N$ in the positive majorant and sum the geometric series with ratio $q_* R < 1$. Substituting the outward safe value 0.700876 gives (10). $\square$

The direct factorwise certificate of RH-264 is stronger at the operational order [9]:

$$\sum_{n \geq 29} \frac{|a_n|}{n} < 0.000026624745, \quad e^{\sum_{n \geq 29} |a_n|/n} - 1 < 0.000026625100. \quad (11)$$

Its reported logarithmic endpoint is more than 6.93 times smaller than the coarse envelope-only endpoint in (10). RH-265 extends this factorwise construction to a certified tail ladder; only the $N = 29$ row is aligned with the currently archived order-28 head, and every row $N \geq 37$ remains a conditional interface rather than a constructed head [10].

Theorem 2.2 (Deterministic target package). *The archived target has a certified analytic tail and certified boundary constant. Its coefficient base is exactly q_* , its logarithmic radius is exactly ρ_* , and its current order-29 tail obeys (11). These conclusions concern the deterministic sequence a_n only and do not imply a moving-cloud coefficient bridge.*

Proof. The boundary and direct tail are (3) and (11). Equations (7)–(8) give the exact root rate and, by Cauchy–Hadamard, the radius ρ_* . None of these statements contains an identification $\tau_{n,s} = a_n$ or a limiting version of that identity, so the bridge does not follow. $\square$

3 The quotient side remains conditional

RH-259 has 23 finite quotient blocks: every twelfth power is contractive, none of the one-step blocks is contractive, and nine archived endpoints are missing [6]. RH-266 proves the exact logical limitation of this evidence: finitely many pointwise contractions do not imply a continuum uniform bound without a modulus or an interval-family enclosure [11]. This does not prove that the actual quotient family is nonuniform.

RH-269 supplies a sufficient continuum theorem [4, 14]. It requires

- (i) Hilbert–Schmidt convergence of the noisy operators;
- (ii) a common finite-rank contour isolating the spectral cluster;
- (iii) a uniform resolvent bound on that contour;
- (iv) a contractive power of the limiting orthogonal quotient.

Together these hypotheses stabilize the Riesz and orthogonal projections and produce uniform RH-246 constants K_m , $\eta_m < 1$, and L_r . The current archive verifies none of the four hypotheses as a continuum theorem. Hence the criterion is exact but inactive, and the uniform quotient tail remains open.

4 Exact updated ledger

component	archived status after RH-269
certified target boundary constant	true; (3)
exact deterministic parity anchor	true; (4)–(5)
direct order-29 deterministic tail	true; (11)
deterministic all-order envelope	true; (7)
sharp deterministic base and radius	true; (8)
legal anchored head	not supplied in audited classes
cloud coefficient bridge	open
uniform quotient tail	open
RH-269 criterion hypotheses	0/4 verified

Theorem 4.1 (Updated complete-certificate ledger). *For Definition 1.1, the archived truth vector is*

$$(H, B, Q, A, M) = (0, 0, 0, 1, 1). \quad (12)$$

Exactly two of five obligations are satisfied. The complete certificate count in the audited route is therefore zero.

Proof. RH-262 certifies $A = M = 1$. RH-263–RH-268 strengthen the target side but do not change the three cloud/quotient entries. The archived head ledger still contains no passing legal anchored head, so $H = 0$; no cloud-to-target coefficient theorem gives $B = 1$; and RH-266–RH-269 leave $Q = 0$, with the sufficient criterion at 0/4 hypotheses. The complete predicate is the conjunction of all five entries, so (12) is incomplete and its audited complete-certificate count is zero. $\square$

Remark 4.2 (Scope of the zero count). The theorem does not exclude a future selector outside the audited head classes, an operator-derived non-idempotent quotient, a cloud bridge, or a continuum proof of the RH-269 hypotheses. It says exactly that no complete certificate is assembled from the presently archived components.

5 Protocol and claim boundary

The companion audit reads the eight structured result files RH-262–RH-269, checks their status fields, theorem boundaries, numerical endpoints, and all forty inherited Gate A–E flags, and writes a deterministic ledger. It does not rerun long operator calculations, refit finite coefficients, or infer uniformity between samples.

The deterministic all-order envelope and sharp base are not moving-cloud theorems. The 27-row parity cross-check is not used as proof of an all-order cloud identity, and the 23 finite contractions are not a uniform quotient certificate. Gates A–E remain false/open. No Hilbert–Polya operator is constructed, no Riemann zeros are identified, no equality with the completed zeta divisor is asserted, and no implication of the Riemann Hypothesis is claimed.

References

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